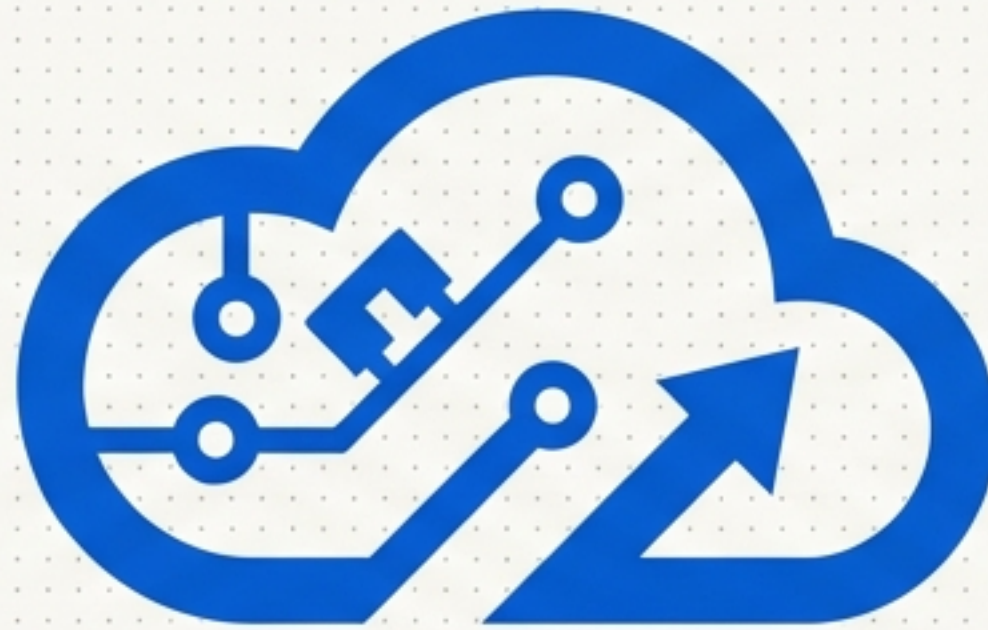




EasyEDA: Panduan Lengkap Desain PCB

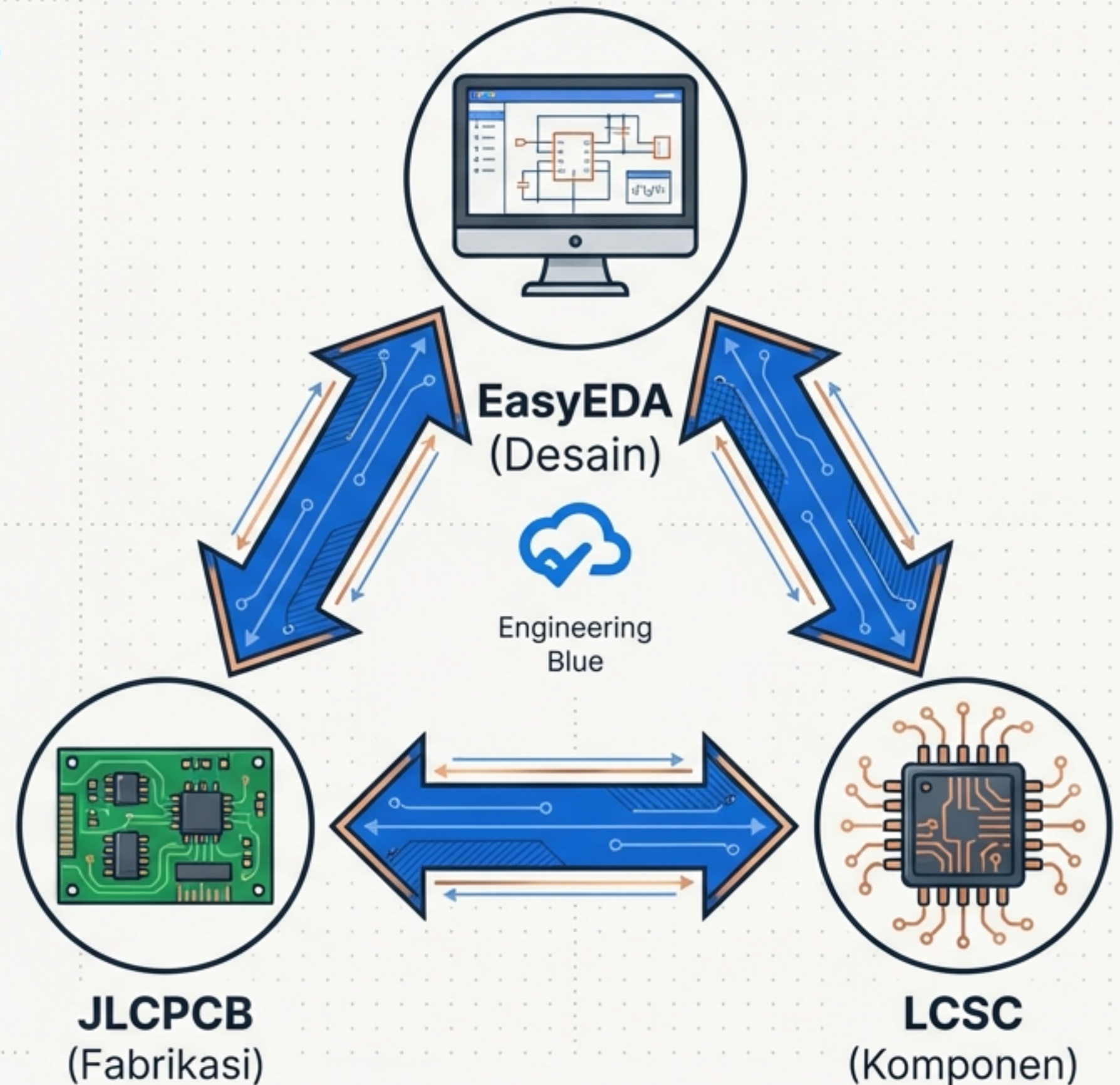
Versi 6.4.32 | 2022

- Web-based EDA (Electronics Design Automation)
- URL Editor: <https://easyeda.com/editor>
- Tersedia desktop client & versi online

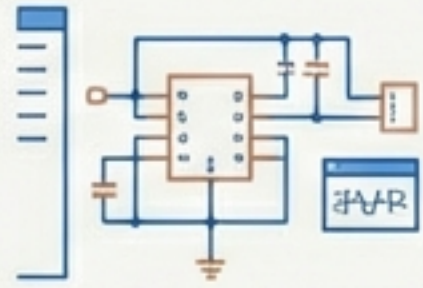
APA ITU EasyEDA?

Definisi

- Tool EDA berbasis cloud, gratis, tanpa instalasi
- Berjalan di browser HTML5 (Chrome/Firefox)
- Platform: Windows, Mac, Linux

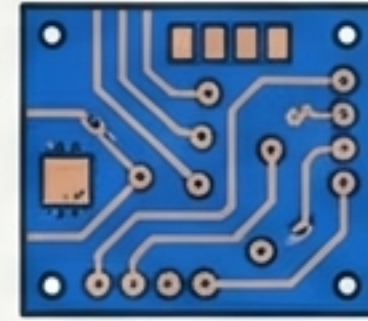


FITUR UTAMA EasyEDA



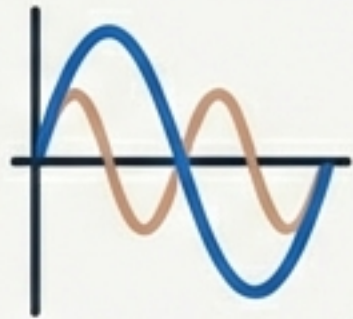
Schematic Capture

Pembuatan skematik profesional



PCB Layout

Layout hingga 6 layer



LTSpice Simulation

Simulasi rangkaian spice



Library 1M+

Simbol & footprint siap pakai



Cloud Collaboration

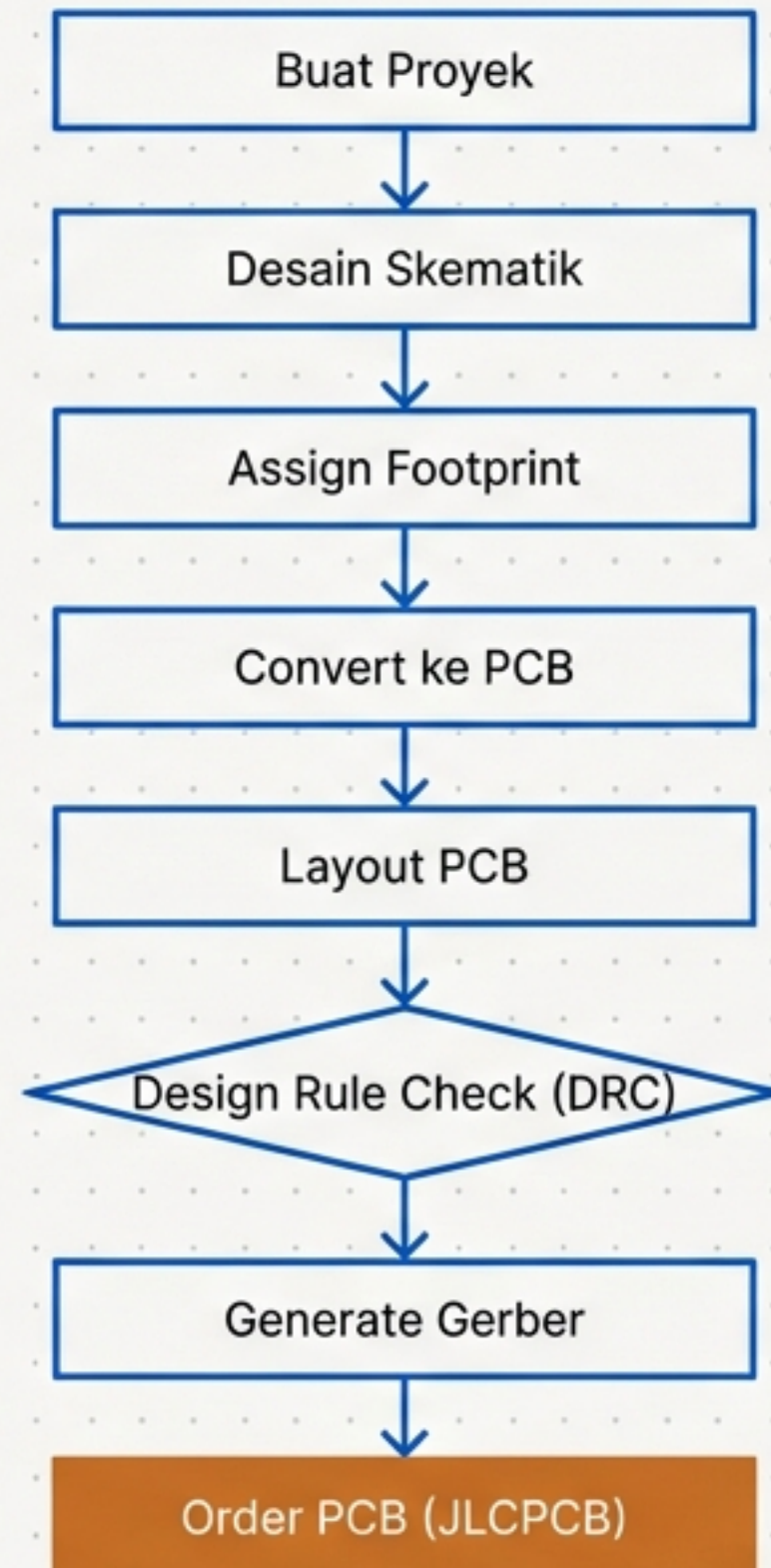
Kerja tim real-time



Gerber Export

File produksi PCB standar

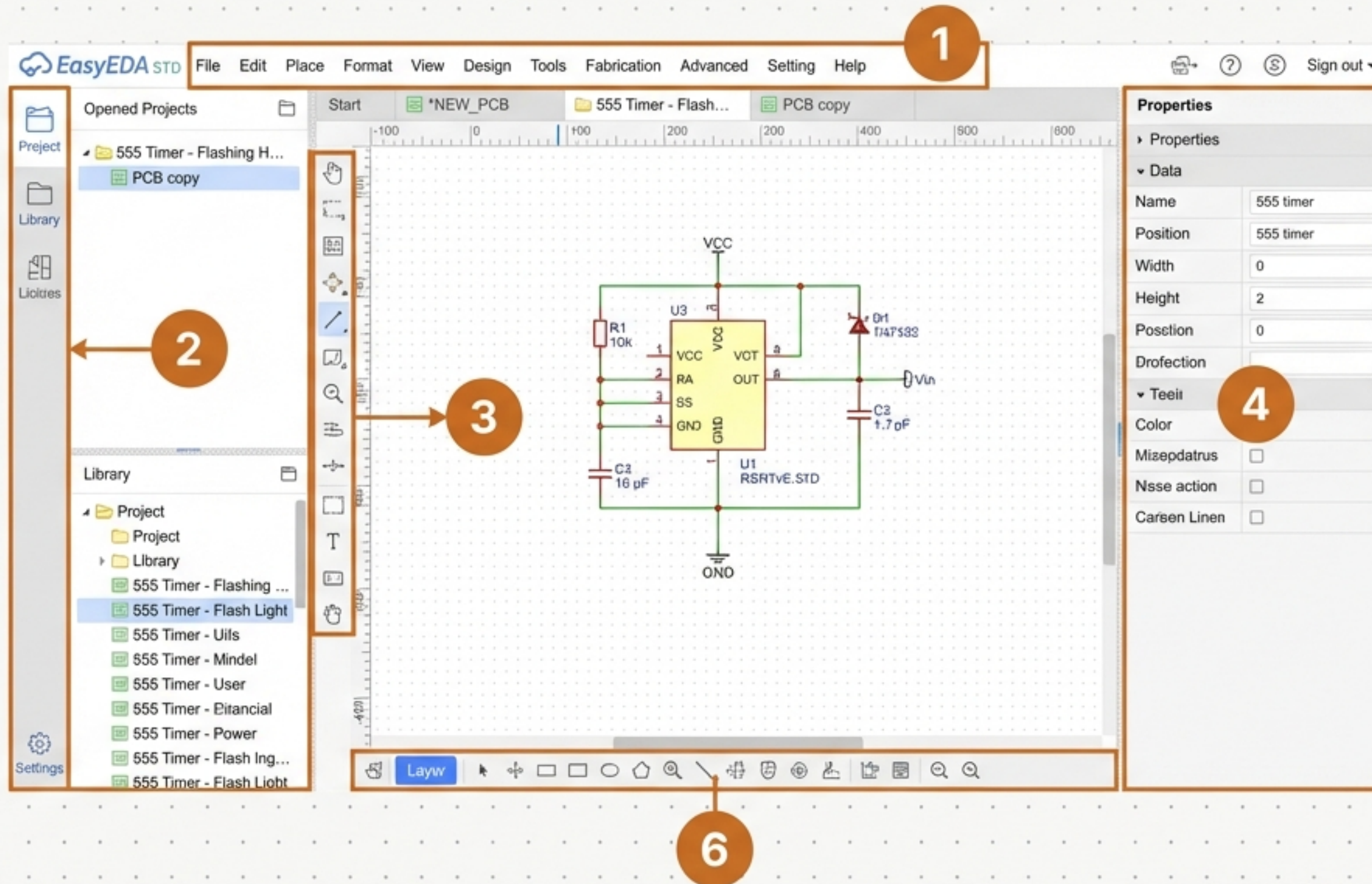
ALUR DESAIN (Design Flow)



PERBANDINGAN DENGAN TOOL LAIN

Aspek	EasyEDA	Altium	KiCAD
Harga	✓ Gratis	Berbayar	Gratis
Platform	✓ Web/Desktop	Windows Only	Multi-OS
Library	✓ 1M+ (LCSC)	Manual	Komunitas
Kolaborasi	✓ Real-time Cloud	Terbatas	Manual (Git)
Fabrikasi	✓ 1-Click JLCPCB	Manual Export	Manual Export

ANTARMUKA PENGGUNA (UI)



1. **Top Menu**
(File, Design, Tools)
2. **Navigation Panel**
(Project, Library)
3. **Canvas**
(Area Desain)
4. **Properties Panel**
(Kanan)
5. **Wiring Tools**
(Toolbar)
6. **Layer Toolbar**
(Bawah)

PANEL NAVIGASI KIRI



Project

} Manajemen file proyek



EElib

} Komponen dasar & simulasi



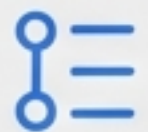
Libraries

} Search 1M+ simbol & footprint



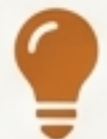
LCSC

} Katalog stok real-time



Design Manager

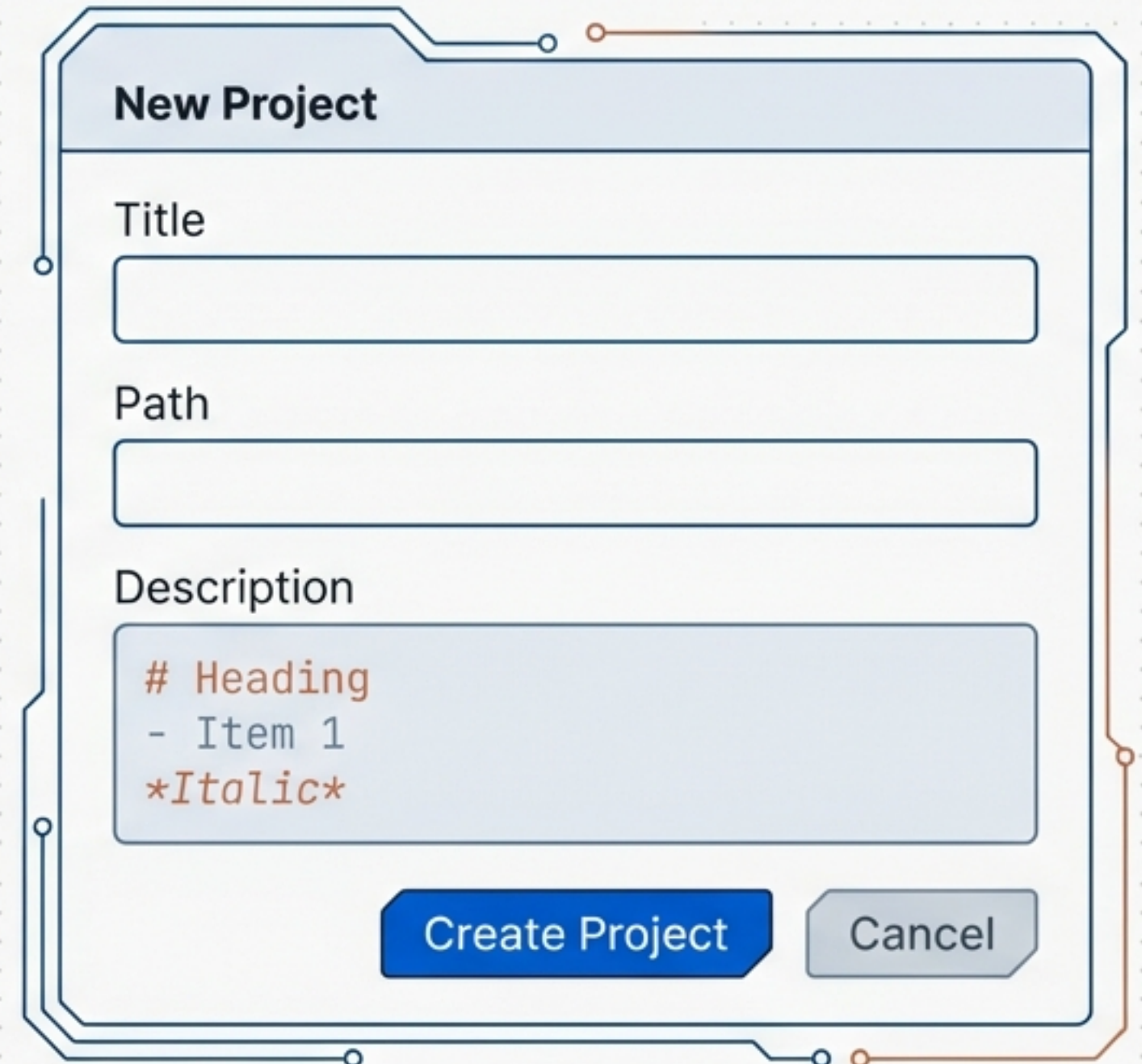
} Navigator Net & DRC Error



Filter: Ketik kata kunci di box filter untuk hasil instan (**Shortcut: Shift+F**)

MEMBUAT PROYEK BARU

1. Menu: File > New > Project
2. Isi **Title** (Nama Proyek)
3. **Path** (URL Unik - permanen)
4. **Description** (Mendukung Markdown)
5. **Visibility**:: Private atau Public



The image shows a 'New Project' dialog box with a light blue header and a white body. It contains three input fields: 'Title', 'Path', and 'Description'. The 'Description' field is a text area with a light blue background and contains the text: `# Heading`, `- Item 1`, and `*Italic*`. At the bottom right, there are two buttons: 'Create Project' (blue) and 'Cancel' (light blue).

New Project

Title

Path

Description

`# Heading`
`- Item 1`
`*Italic*`

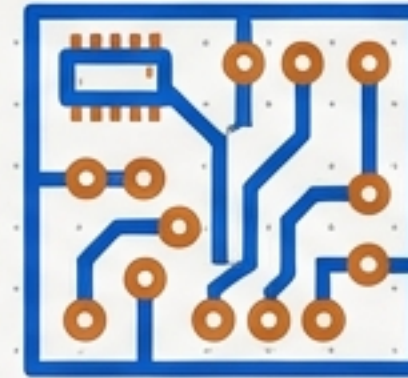
Create Project Cancel

MEMBUAT FILE DI PROYEK

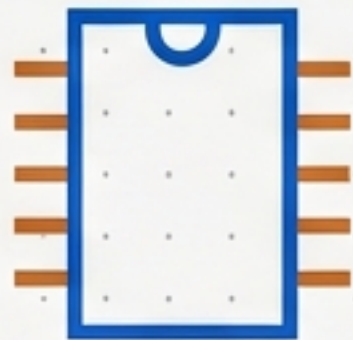
`File > New > ...`



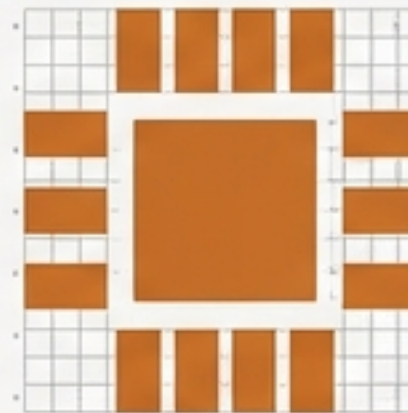
Schematic
Logika Rangkaian



PCB
Layout Fisik Board



Symbol
Gambar untuk Skematik



Footprint
Landasan Tembaga PCB

PENGATURAN CANVAS SKEMATIK

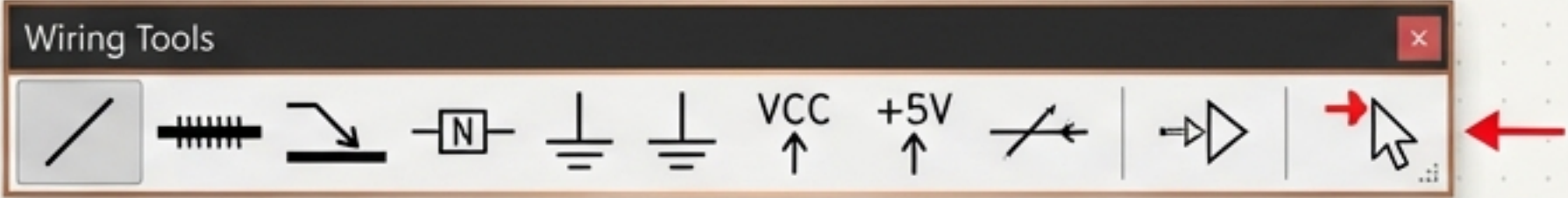
- **Grid Size:** Rekomendasi 10, 20, atau 100 pixel
- **Snap = Yes:** WAJIB AKTIF (Mencegah pin floating)
- **ALT Snap:** Tahan ALT untuk gerakan halus
- **Canvas Size:** Preset A4/A3 atau Custom

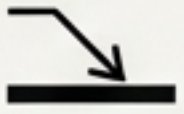
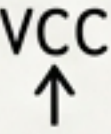
The image shows a 'Canvas Attributes' dialog box with the following settings:

Canvas Attributes	
Background	#FFFFFF
Visible Grid	Yes
Grid Color	#CCCCCC
Grid Style	line
Grid Size	5
Snap	Yes

Arrows from the list on the left point to the 'Visible Grid', 'Grid Size', and 'Snap' settings in the dialog box. The 'Snap' setting is highlighted with an orange border.

WIRING TOOLS — PENGANTAR



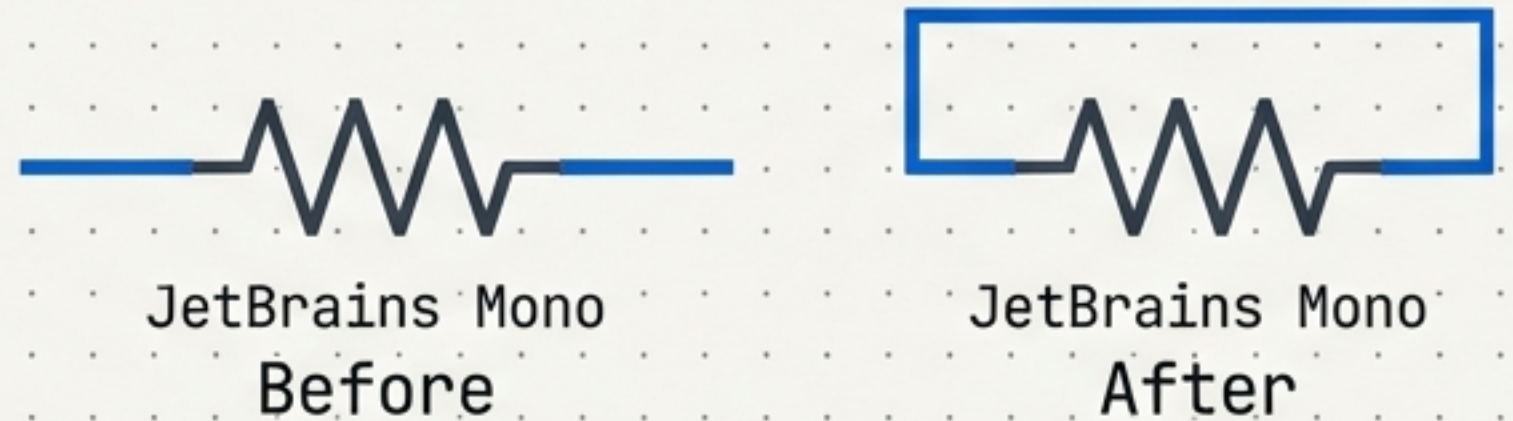
ICON (IKON)	KEY (PINTASAN)	FUNCTION (FUNGSI)
	W	Wire (Kabel Penghubung)
	B	Bus (Jalur Sinyal Grup)
	U	Bus Entry
	N	Net Label (Nama Jaringan)
	Ctrl+G	Ground Flag
	Ctrl+Q	VCC Power Flag

MENGGAMBAR WIRE (SAMBUNGAN)

Cara Menggambar

- 1. Tekan hotkey `W`.
- 2. Klik ujung pin komponen (titik abu-abu).
- 3. Klik titik tujuan.

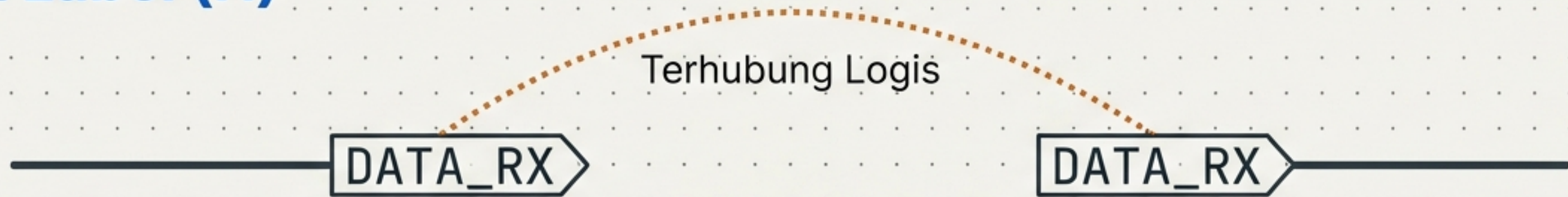
Fitur Auto-Connect



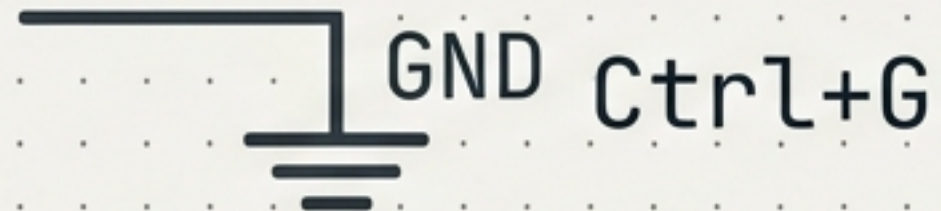
⚠ Jangan gunakan "Drawing Line" untuk koneksi listrik! Gunakan hanya "Wire" tool.

NET LABEL & NET FLAG

Net Label (N)



Net Flag



Rule Check

Good

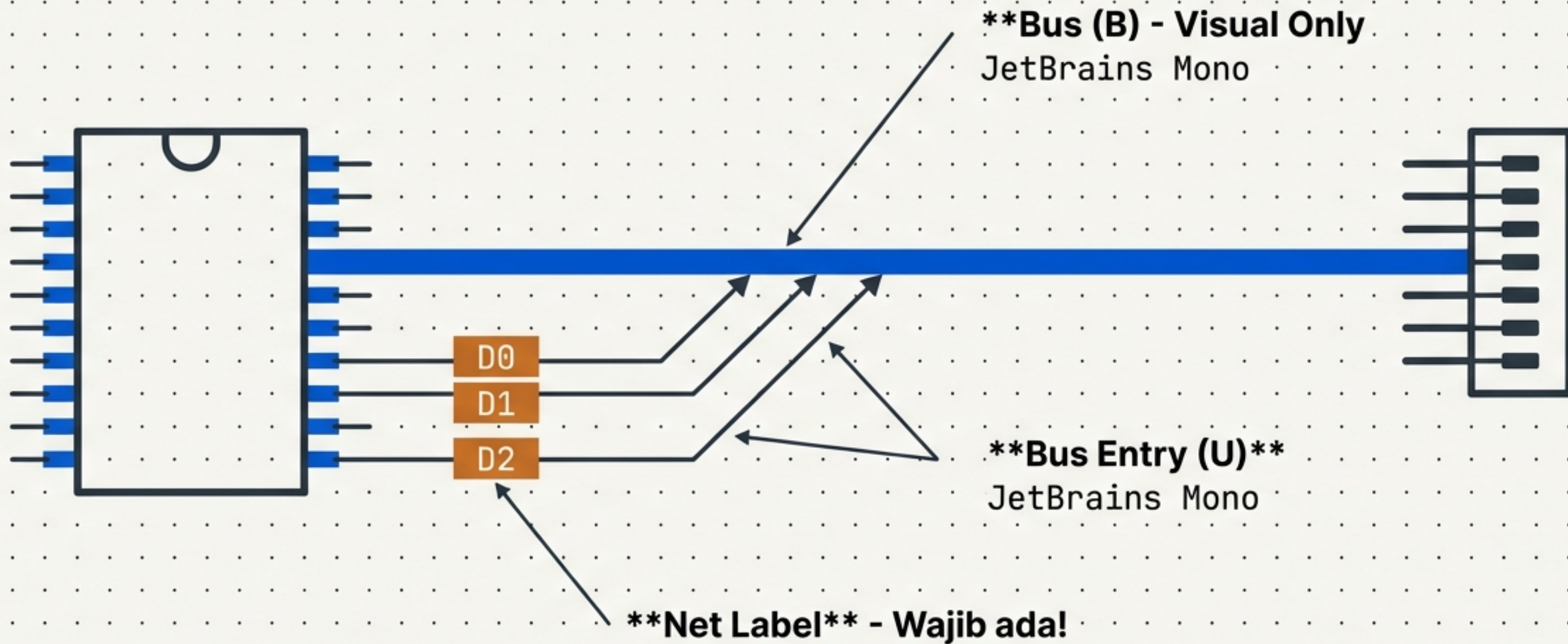


Bad

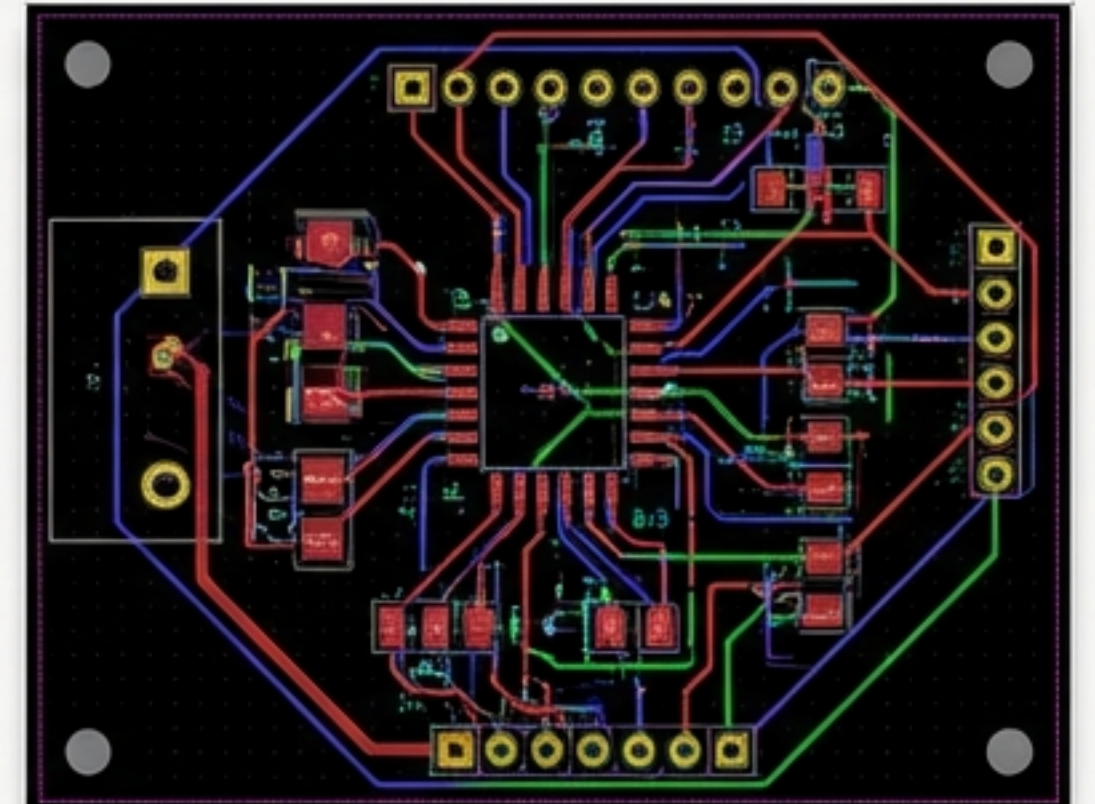
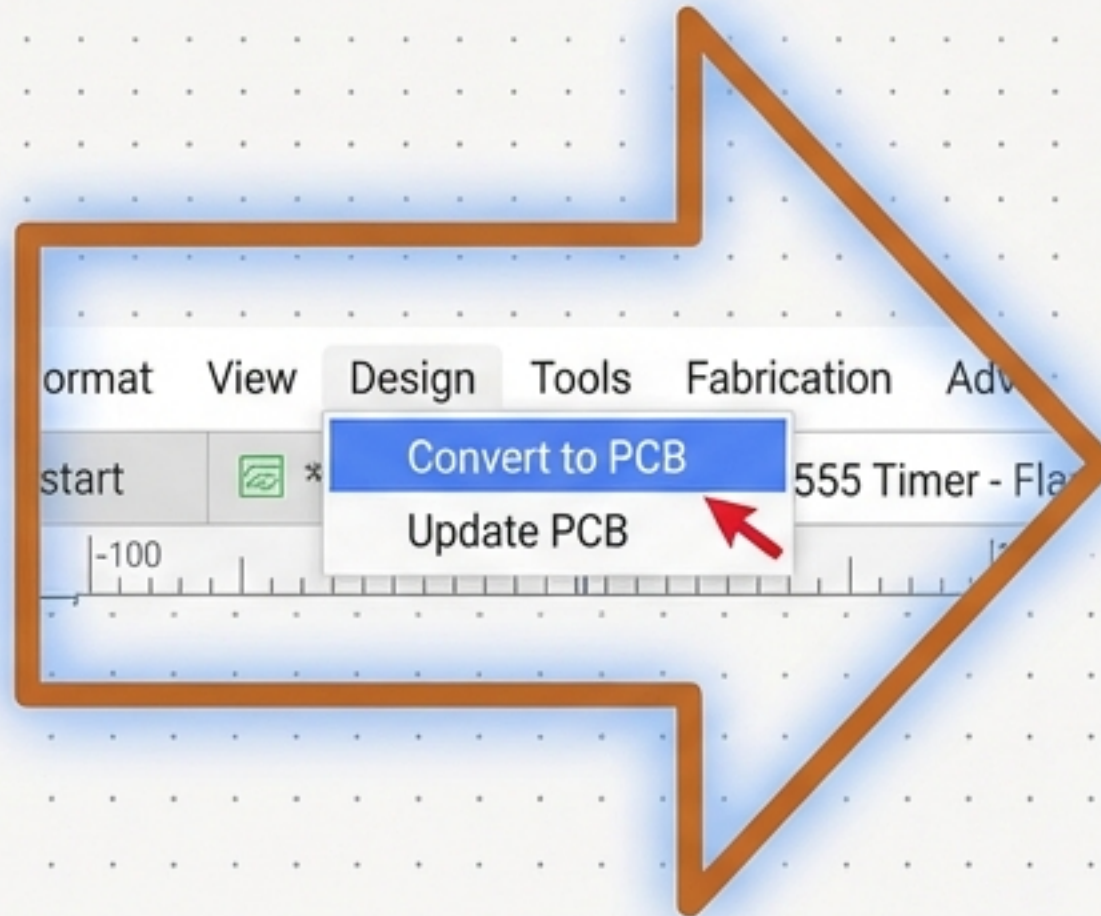
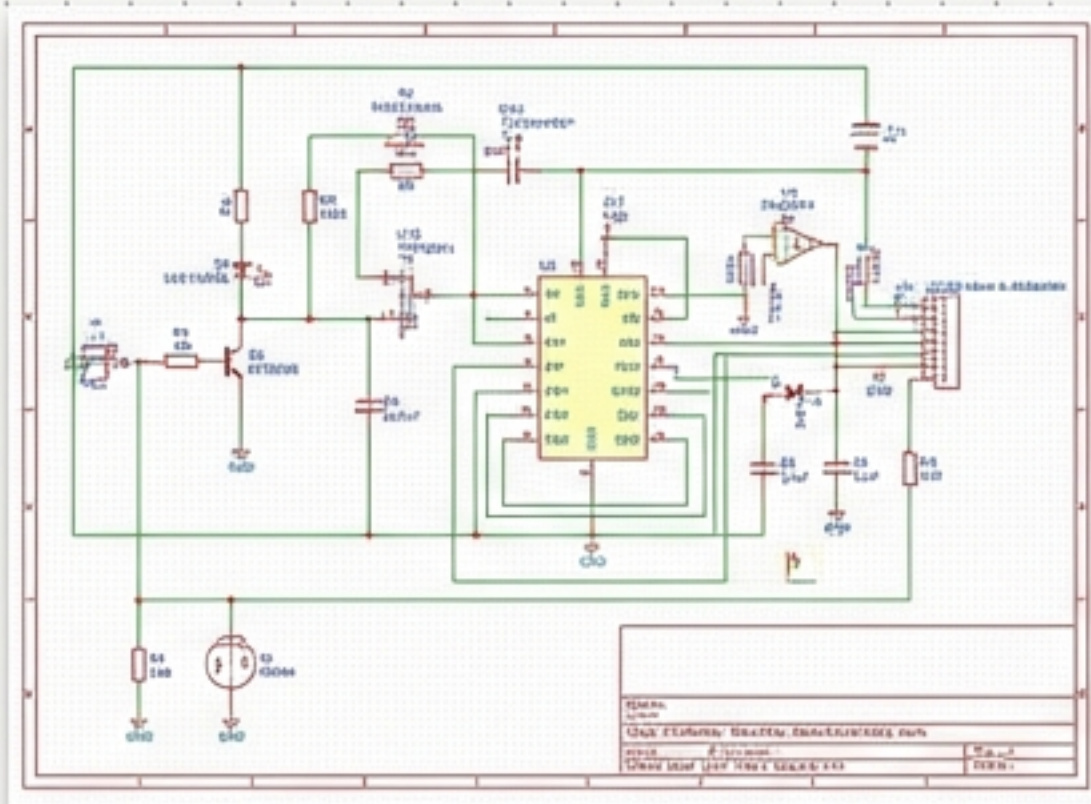


Label tidak terhubung ke pin atau wire!

BUS & BUS ENTRY



DARI SKEMATIK KE PCB



- 1. Validasi (ERC Check) ✓ 
- 2. Convert to PCB  
- 3. Layout & Routing   
- 4. Order JLCPCB 

**Siap mencetak
desain Anda?**